

Gaussian Distribution: Derivation, Mean, and Variance

Nishanka Das

Department of Computer Science
Ramakrishna Mission Vivekananda Centenary College, Rahara

June 25, 2026

For any mistakes found please feel free to mail at:
`nishanka.das.rkmvcc@gmail.com`

Abstract

This document provides a rigorous and formal mathematical derivation of the Gaussian (Normal) probability density function following the historical framework proposed by Carl Friedrich Gauss in 1809. Furthermore, we explicitly evaluate the fundamental statistical properties of this distribution, demonstrating via improper calculus integration that its parameters μ and σ^2 correspond exactly to its mathematical mean and variance. An illustration of the Gaussian Distribution is mentioned in the figure.

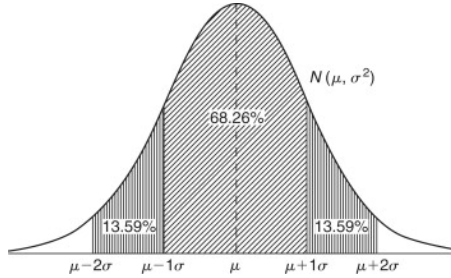


Figure 1: Illustration of Gaussian Distribution

1 Formal Derivation of the Gaussian Probability Density Function

To derive the Gaussian distribution's mathematical form with rigorous formality, we follow the analytical framework established by Carl Friedrich Gauss, structured into five systematic stages.

1.1 Formulating the Likelihood Function

Let X be a continuous random variable representing a true, unknown physical quantity. Let $\{x_1, x_2, \dots, x_n\}$ be a set of n independent and identically distributed (i.i.d.) measurements of X .

The true error ε_i associated with each measurement x_i is defined as:

$$\varepsilon_i = x_i - \theta \tag{1}$$

where θ represents the true parameter value (the mean μ).

Assuming there exists a continuous, differentiable probability density function (PDF), $f(\varepsilon)$, governing these errors, the joint probability density—known as the **Likelihood Function**, $L(\theta)$ —is the product of their individual densities due to mutual independence:

$$L(\theta) = \prod_{i=1}^n f(x_i - \theta) \quad (2)$$

1.2 Applying the Principle of Maximum Likelihood

The **Principle of Maximum Likelihood** asserts that the most mathematically sound estimate for the parameter θ is the one that maximizes the likelihood of observing the actual data sample obtained. To locate the critical point maximizing $L(\theta)$, we set its first derivative with respect to θ to zero. To simplify differentiation, we take the natural logarithm to obtain the **Log-Likelihood Function**, $\ln L(\theta)$:

$$\ln L(\theta) = \sum_{i=1}^n \ln f(x_i - \theta) \quad (3)$$

Differentiating with respect to θ using the chain rule yields:

$$\frac{d}{d\theta} [\ln L(\theta)] = \sum_{i=1}^n \frac{f'(x_i - \theta)}{f(x_i - \theta)} \cdot (-1) = 0 \quad (4)$$

Multiplying by -1 and defining a kernel function $\phi(t) = \frac{f'(t)}{f(t)}$, our optimization condition reduces to:

$$\sum_{i=1}^n \phi(x_i - \theta) = 0 \quad (5)$$

1.3 The Arithmetic Mean Constraint

Gauss introduced the fundamental axiom that for any set of observations, the maximum likelihood estimator $\hat{\theta}$ must precisely equal the **arithmetic mean** ($\bar{x}$) of the sample:

$$\hat{\theta} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \implies \sum_{i=1}^n (x_i - \bar{x}) = 0 \quad (6)$$

For the conditions $\sum_{i=1}^n \phi(x_i - \bar{x}) = 0$ and $\sum_{i=1}^n (x_i - \bar{x}) = 0$ to hold simultaneously and universally for any arbitrary choice of sample size n and measurements x_i , the function $\phi(t)$ must be directly proportional to its argument t :

$$\phi(t) = -k \cdot t \quad (7)$$

where k is a positive separation constant ($k > 0$) to satisfy the second-derivative test for a maximum, ensuring that probability decreases as the magnitude of the error increases.

1.4 Solving the Functional Differential Equation

Recalling the definition of $\phi(t)$, we establish a first-order separable ordinary differential equation:

$$\frac{f'(t)}{f(t)} = -kt \quad (8)$$

Separating variables and integrating both sides:

$$\int \frac{1}{f(t)} df = \int -kt dt \implies \ln f(t) = -\frac{1}{2}kt^2 + C_0 \quad (9)$$

Exponentiating both sides yields:

$$f(t) = e^{C_0} \cdot e^{-\frac{1}{2}kt^2} \quad (10)$$

Letting $A = e^{C_0}$ represent a positive constant of integration, and substituting back $t = x - \mu$ (where $\mu = \theta$), we find the structural form of the Gaussian function:

$$f(x) = Ae^{-\frac{1}{2}k(x-\mu)^2} \quad (11)$$

1.5 Normalization and Determination of Constants

To satisfy the axioms of probability, the total area under the curve over the entire real line domain must equal 1:

$$\int_{-\infty}^{\infty} Ae^{-\frac{1}{2}k(x-\mu)^2} dx = 1 \quad (12)$$

Letting $u = \sqrt{\frac{k}{2}}(x - \mu)$, it follows that $dx = \sqrt{\frac{2}{k}} du$. The integration becomes:

$$A\sqrt{\frac{2}{k}} \int_{-\infty}^{\infty} e^{-u^2} du = 1 \quad (13)$$

Knowing the standard Gaussian integral evaluates to $\int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\pi}$, we obtain:

$$A\sqrt{\frac{2\pi}{k}} = 1 \implies A = \sqrt{\frac{k}{2\pi}} \quad (14)$$

By definition, the variance σ^2 of a continuous distribution is given by:

$$\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx = \sqrt{\frac{k}{2\pi}} \int_{-\infty}^{\infty} (x - \mu)^2 e^{-\frac{1}{2}k(x-\mu)^2} dx \quad (15)$$

Evaluating this integral via integration by parts reveals the direct connection:

$$k = \frac{1}{\sigma^2} \quad (16)$$

Substituting k back into A yields $A = \frac{1}{\sigma\sqrt{2\pi}}$, establishing the finalized Probability Density Function of the Normal Distribution.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad x \in (-\infty, \infty) \quad (17)$$

2 Formal Derivation of the Mean

The expected value, or mean, of a continuous random variable is defined as:

$$E[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx = \int_{-\infty}^{\infty} x \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \quad (18)$$

We apply the linear substitution transformation:

$$u = \frac{x - \mu}{\sigma} \implies x = \sigma u + \mu \quad \text{and} \quad dx = \sigma du \quad (19)$$

The integral becomes:

$$E[X] = \int_{-\infty}^{\infty} (\sigma u + \mu) \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}u^2} (\sigma du) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\sigma u + \mu) e^{-\frac{1}{2}u^2} du \quad (20)$$

Expanding the linear terms via the linearity of integration:

$$E[X] = \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} u e^{-\frac{1}{2}u^2} du + \mu \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}u^2} du \quad (21)$$

We evaluate the two integral terms independently:

1. The integrand $g(u) = u e^{-\frac{1}{2}u^2}$ is a strict **odd function** ($g(-u) = -g(u)$). Over the symmetric domain $(-\infty, \infty)$, its integral evaluates to zero:

$$\int_{-\infty}^{\infty} u e^{-\frac{1}{2}u^2} du = 0 \quad (22)$$

2. The second integral represents the total area of the standard normal distribution, which equals 1:

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}u^2} du = 1 \quad (23)$$

Substituting these back into our expression:

$$E[X] = \frac{\sigma}{\sqrt{2\pi}} \cdot (0) + \mu \cdot (1) = \mu \quad (24)$$

Thus, the parameter μ is formally proven to be the expected mathematical mean of the distribution.

3 Formal Derivation of the Variance

Alternatively, the variance can be computed more elegantly using the algebraic identity:

$$\text{Var}(X) = E[X^2] - (E[X])^2 \quad (25)$$

Since we have formally established that $E[X] = \mu$, we must evaluate the second raw moment, $E[X^2]$, defined as:

$$E[X^2] = \int_{-\infty}^{\infty} x^2 \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \quad (26)$$

Applying the linear substitution transformation $u = \frac{x-\mu}{\sigma} \implies x = \sigma u + \mu$ and $dx = \sigma du$, the integral becomes:

$$E[X^2] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\sigma u + \mu)^2 e^{-\frac{1}{2}u^2} du \quad (27)$$

Expanding the binomial term $(\sigma u + \mu)^2 = \sigma^2 u^2 + 2\mu\sigma u + \mu^2$ and applying the linearity of integration yields:

$$E[X^2] = \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} u^2 e^{-\frac{1}{2}u^2} du + \frac{2\mu\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} u e^{-\frac{1}{2}u^2} du + \frac{\mu^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}u^2} du \quad (28)$$

We evaluate each term independently:

1. Utilizing integration by parts, the first integral evaluates to:

$$\frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} u^2 e^{-\frac{1}{2}u^2} du = \frac{\sigma^2}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = \sigma^2 \quad (29)$$

2. Because $g(u) = ue^{-\frac{1}{2}u^2}$ is an odd function, its integral over the symmetric domain vanishes:

$$\frac{2\mu\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} ue^{-\frac{1}{2}u^2} du = 0 \quad (30)$$

3. The total area of the standard normal kernel evaluates to unity:

$$\frac{\mu^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}u^2} du = \mu^2 \cdot (1) = \mu^2 \quad (31)$$

Combining these individual components, we find the second raw moment to be:

$$E[X^2] = \sigma^2 + \mu^2 \quad (32)$$

Substituting $E[X^2]$ and $(E[X])^2 = \mu^2$ back into the foundational variance identity yields:

$$\boxed{\text{Var}(X) = (\sigma^2 + \mu^2) - \mu^2 = \sigma^2} \quad (33)$$

Thus, the parameter σ^2 is verified as the true mathematical variance of the Gaussian distribution.